Taekhyun Park (박택현)

pthpark1@pusan.ac.kr | Github
Personal Website | Google Scholar | LinkedIn | Hugging Face

Busan, South Korea

Research Interests

- **Novel architectures:** JEPA, deep supervision, state space models, and looped LLMs
- **Sequence modeling:** time series, language, video, world, and event log modeling
- **LLM systems:** process mining with LLMs, RLVR, and agent-based systems

Education

Pusan National University — Mar. 2025 – Feb. 2027
M.S. in Data Science, Graduate School of Data Science — Busan, South Korea
– Advisor: Prof. Hyerim Bae

Korea Maritime & Ocean University — Mar. 2019 – Feb. 2025
B.S. in Control Engineering — Busan, South Korea
– Completed military service in the Republic of Korea Army, Oct. 2020 – Apr. 2022

Projects

AI기반 빅데이터 통계시스템 구축 및 체인포털 서비스 개선 사업 — June. 2026 – Dec. 2026
Funded by Busan Port Authority(BPA)

해상교통 환경 특화 대형행동모형 개발을 위한 에이전트 기술 개발 — June. 2026 – Dec. 2026
Funded by Korea Research Institute of Ships & Ocean Engineering (KRISO)

프로세스 마이닝을 활용한 고객 사용 패턴 기반 개인화 스마트 운전 로직 개발 — Dec. 2025 – Nov. 2026
Funded by LG Electronics

이상치 탐지 배원시뮬레이터 개발 — Oct. 2025 – Dec. 2025
Funded by Samsung Heavy Industries

동적 해양사고 시나리오 도출 및 실시간 사고위험 통합지표 기반 조기경보 시스템 개발 — Aug. 2025 – Jan. 2026
Funded by Pusan National University and RISE

인간 중심 - 탄소 중립 글로벌 공급망 연구센터 — Mar. 2025 – Feb. 2027
Funded by the National Research Foundation of Korea

Publications

Journal & Inproceedings

[I]Design and Implementation of an LLM-Based Maritime Risk Scenario Generation System for Vessel Traffic Services — IEEE CASE, 2026
– **Taekhyun Park***, Sangmin Jo, Dohee Kim, and Hyerim Bae

[J]LURKER: Language Models for Unveiling Maritime Risks via Knowledge-Driven Event Forecasting and Reasoning — ICIC Express Letters
– Seongmoon Hong*, **Taekhyun Park**, Dohee Kim, Sangmin Jo, and Hyerim Bae

[J]Identifying Severity of Maritime Accidents Based on Weighted Kernel Density Estimation — ICIC Express Letters
– Sangmin Jo*, Dohee Kim, Seongmoon Hong, **Taekhyun Park**, and Hyerim Bae

[I]JustDense: Just using Dense instead of Sequence Mixer for Time Series Analysis — IEEE BigData 2025
– **Taekhyun Park***, Yongjae Lee*, Daesan Park*, Dohee Kim, and Hyerim Bae
– Code

[J]LA-RCS: LLM-agent-based Robot Control System — Sensors and Materials
– **Taekhyun Park***, Young-Jun Choi, Seung-Hoon Shin, Chang-Eun Lee, and Kwangil Lee
– Project Page

Last updated: 2026.08.10

Preprints

Variable-Horizon Workforce Demand Forecasting with an Aggregate Demand Constraint for Construction Workforce Planning — arXiv, 2026
– Hanbyeol Park*, Jaehyeon Heo*, **Taekhyun Park**, Minseong Kim, and Hyerim Bae

Trie-Constrained Token Prediction with Hierarchy-Aware Semantic Alignment for HS Code Prediction — arXiv, 2026
– Minseop Kim*, **Taekhyun Park**, Kikun Park, and Hyerim Bae

LoopUS: Recasting Pretrained LLMs into Looped Latent Refinement Models — arXiv, 2026
– **Taekhyun Park***, Yongjae Lee, Dohee Kim, and Hyerim Bae
– Project Page

Rewarding Structural Conformance of Reasoning using Process Mining — arXiv, 2025
– Yongjae Lee*, **Taekhyun Park***, and Hyerim Bae
– Code

Generative AI and Machine Learning Collaboration for Container Dwell Time Prediction via Data Standardization — arXiv, 2026
– Minseop Kim*, Takhyeong Kim, **Taekhyun Park**, Hanbyeol Park, and Hyerim Bae

Application of Large Language Models for Container Throughput Forecasting: Incorporating Contextual Information in Port Logistics — arXiv, 2026
– Minseop Kim*, Jaeeun Kwon, Hanbyeol Park, Kikun Park, **Taekhyun Park**, and Hyerim Bae

Selected Presentations

Qwen: Global AI Strategy and Technical Ecosystem — SCSC LLM WG Seminar, Jul. 2026

Introduction of JEPA — Weekly Lab Seminar, Jun. 2026

Recursive Model — Weekly Lab Seminar, Feb. 2026

BitNet Overview — Freshman Study, Feb. 2025

From RNN to Mamba2: Solving Online Function Approximation Problems in Post-LLM Models — Freshman Study, Feb. 2025

Internship

DeepLogicChain — Jan. 2024 – Mar. 2024
Intern — Seoul, South Korea

Teaching Experience

Pusan National University — Mar. 2025 – Jun. 2026
Teaching Assistant for Data Structures and Algorithms (Undergraduate Course) — Busan, South Korea

Honors & Awards

부산테크노파크 원장상 — 2025
실시간 데이터 및 해양 규제 융합형 LLM 기반 사고 대응 지원 시스템 — K-해양 AI 챌린지

레인보우브레인상 — 2023
RAG 및 피드백 기반 AI 기법을 활용한 학교 게시판 검색 AI 시스템 — KMOU 캡스톤디자인

대상 — 2022
이미지 생성 AI 학습을 위한 P2P 기반 데이터셋 구축 및 AI 소유권 가상자산화 플랫폼 — KMOU AI 아이디어 챌린지

경남테크노파크 원장상 — 2022
블록체인 기술을 활용한 글쓰기 하이퍼 플랫폼 — AI & 블록체인 아이디어 경진대회

Tech stack

Languages: Python, R, Go, C/C++, Ruby, VHDL, MATLAB, Solidity, Elixir
Tools: CUDA, Triton, PyTorch